Advances and Applications of Cuckoo Search Algorithm in Scheduling Problems

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ABSTRACT

Scheduling of jobs is a complex combinatorial problem met in many real world situations and in most scenarios these problems are NP-hard. Though there are a lot of solution approaches (heuristics and metaheuristic algorithms) for these kinds of problems,a recently developed nature inspired metaheuristic algorithm called Cuckoo Search algorithm is widely gaining attention. This paper presents a detailed outline of Cuckoo Search algorithm followed by its advances and applications in scheduling problems. The paper is then concluded by giving future research directions.

Keywords: *Cuckoo Search (CS) Algorithm, Lévy Flights, Metaheuristics, Optimization, Scheduling.*

I. INTRODUCTION

Scheduling is considered to be a process of assigning resources over time to perform a specific collection of tasks [1]. It is a type of decision making process and plays a vital role for the growth of any kind of industries. Scheduling problems are of different types and were addressed in the literature having different performance measures. Single machine scheduling, parallel machine scheduling, flow shop scheduling, hybrid flow shop (HFS) scheduling and job shop scheduling are some of the important types of scheduling environments.

Scheduling problems are often combinatorial optimization problems and in most cases, they are NP-hard for even a single optimization criterion. Therefore, solution approaches are classified into [2]: (1) finding optimal solution using exact algorithms such as the branch and bound method and the mixed integer linear programming method which requires ample computation time; (2) finding a near optimal solution using heuristic methods which can obtain a solution in a reasonable computational time. Heuristics algorithms can be constructive algorithmsor improvement heuristic algorithms (metaheuristic techniques) such as Genetic Algorithms (GA), Simulated Annealing (SA) and Particle Swarm Optimization (PSO).

Recently developed, a nature inspired metaheuristic algorithm called Cuckoo Search (CS) is widely gaining attention because of its ability to undergo fast local search and finding global optimum for given problems. Yang and Deb (2009) [3] developed this algorithm inspired from the obligate brood parasitic behaviour of some cuckoo species in combination with the Lévy flight behaviour of some birds and fruit flies. CS algorithm was validated on various test functions and the result comparison have proved that this algorithm is very effective and outperforms other algorithms such as GA and PSO. Furthermore, CS algorithm had been applied on engineering optimization [4] by showing its applicability in engineering problems such as spring design optimization and welded beam design. Since it has few parameters to be

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tuned compared to other metaheuristics, CS algorithm can be widely applied in many engineering problems easily and effectively. In this paper outline of CS algorithm approach has been discussed in detail followed by its application and advances in scheduling problems.

The rest of the paper is organised as follows: The CS algorithm is explained in Section II, its advancements and applications in scheduling problems are presented in Section III and Conclusions and future research opportunities will be discussed in Section IV.

II. CUCKOO SEARCH ALGORITHM

Yang and Deb (2009) [3]developed a nature inspired metaheuristic algorithm called the Cuckoo Search (CS) algorithm,enthused from the obligate brood parasitic behaviour of some cuckoospecies in combination with the Lévy flight behaviour of some birdsand fruit flies in nature. In order to describe the CS algorithm more clearly the breeding behaviour and the Lévy flightswill be discussed beforehand and afterward, the basic ideas and the steps of the CS algorithm will be outlined.

2.1 Cuckoo Breeding Behaviour

The beautiful sound of cuckoos is astonishing and it's the first impression that comes in our mind about the cuckoos. But apart from their sound, they have a remarkable aggressive reproductive strategy. This is known as brood parasitism. Brood parasitism are of two types: non-obligatory and obligatory. In non-obligatory brood parasitism, cuckoo lays eggs in the nest of same species or in their own nests. But quite a number of cuckoo species engage in the obligate brood parasitism by laying their eggs in the nests of other host birds (often other species) and thereby removing host's eggs to increase the hatching probability of their own eggs. At times the host birds may engage direct fight with intruder cuckoos. If the host bird learns about the eggs of the intruder, they may either throw out the alien eggs or may leave its nest and rebuild the nest elsewhere. Some parasitic cuckoo species have evolved specializing in imitating the colour and the shape of the eggs of chosen host birds which in turn reduces the probability of their eggs being abandoned and increases their reproductivity. In addition, the time of egg laying of some cuckoo species is also remarkable. The cuckoos often select a nest where the host bird just laid its own eggs. It is also found that cuckoo eggs hatch slightly earlier than their host eggs. Once the first cuckoo chick is hatched, the first instinct action it will take is to expel the host eggs by blindly pushing the eggs out of the nest, which increases the cuckoo chick's share of foodprovided by its host bird. From studies, it has also been understood that cuckoo chick can mimic the sound of host chicks to obtain more food from the host mother.

2.2 Lévy Flight

Generally, animals search for food in a random or quasi-random manner. In nature, foraging path of any animal is effectively a random walk as the next move depends on the current location or state and the transition probability to the next location. Which direction for choosing depends implicitly on a probability which can be modelled mathematically. Various investigations have shown that the flight behaviour of many animals and insects has demonstrated the typical characteristics of Lévy flights [5]. Thus a Lévy Flight can be thought of as a random walk where the step size has a Lévy tailed probability distribution. The name Lévy Flight came after the French mathematician Paul PierreLévy. The term Lévy Flight was coined by Benoit Mandelbrot who used specific definition for the distribution of the step size.Lévyflight is used in cuckoo search for generating new nests or solutions after each step. Ultimately, such behaviour has been applied to optimization and optimal search, and preliminary results show its promising capability.

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2.3 Cuckoo Search Algorithm

For simplicity in describing Cuckoo Search algorithm [3] following three idealized rules/assumptions are described:

1. Each cuckoo lays one egg at a time, and dumps it in a randomly chosen nest;
2. The best nests with high quality of eggs (solutions) will carry over to the next generations;
3. The number of available host nests is fixed, and a host can discover an alien egg with a probability $P_a \in [0, 1]$. In this case, the host bird can either throw the egg away or abandon the nest so as to build a completely new nest in a new location.

For simplicity, this last assumption can be approximated by a fraction P_a of the nnests being replaced by new nests (with new random solutions at new locations). For a maximization problem, the quality or fitness of a solution can simply be proportional to the objective function. For easiness, the following simple representationscan be used:

- Each egg in a nest represents a solution, and a cuckoo egg represent a new solution,
- The aim is to use the new and potentially better solutions (cuckoos) to replace a not-so-good solution in the nests,
- In the simplest form, each nest has one egg. The algorithm can be extended to more complicated cases in which each nest has multiple eggs representing a set of solutions.

Based on these rules, the basic steps of the Cuckoo Search (CS) can be summarised as the pseudo code shown in Fig. 2.1

begin
Objective function f(x), x = (x_1,....., x_d)
Generate initial population of
nhost nests x_i (i = 1, 2, ..., n)
while (t <*MaxGeneration) or (stop criterion)*
Get a cuckoo randomly by Lévy flights
evaluate its quality/fitness F_i
Choose a nest among n(say, j*) randomly*
if ($F_i > F_j$),
replace j *by the new solution;*
end if
A fraction (P_a) *of worse nests*
are abandoned and new ones are built;
Keep the best solutions
(or nests with quality solutions);
Rank the solutions and find the current best
end while
Post process results and visualization
end

Fig. 2.1 Pseudo Code of the Cuckoo Search (CS) [3]

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When generating new solutions $x^{(t+1)}$for, say cuckoo *i*, a Lévy flight is performed:

$$x_i^{(t+1)} = x_i^{(t)} + \alpha \oplus \text{Lévy } (\lambda) \quad (1)$$

where α >0 is the step size which should be related to the scales of the problem of interest. In most cases, we can use α = *O* (1). The above equation is the stochastic equation for random walk. Generally, a random walk is a Markov chain whose next status/location only depends on the current location (the first term in the above equation) and the transition probability (the second term). The product ⊕means entry wise multiplications. Lévy flights essentially provide a random walk while their random steps are drawn from a Lévy distribution for large steps:

$$\text{Lévy} \sim u = t^{-\lambda}, (1 < \lambda \leq 3) \quad (2)$$

which has an infinite variance with an infinite mean. Here the consecutive jumps/steps of a cuckoo essentially form a random walk process which obeys a power-law step-length distribution with a heavy tail.

III. SCHEDULING BY CUCKOO SEARCH

This section reviews the application of CS algorithmand its advancements in scheduling problems.

The permutationflow shop scheduling problem (PFSSP) is the simplifiedversion of the flow shop scheduling problem, and has been proved to be non-deterministic-polynomial-time (NP)-hard. A Hybrid Cuckoo Search (HCS) algorithm was proposed by Li and Yin (2013) [6] to solve these kinds of scheduling problem. In the standard CS approach, they introduced firstly a LRV rule to convert the continuous encoding in CS to a discrete job permutation and then NEH heuristic was used prior to CS to initialise the population. And finally a fast local search is proposed to enhance the local exploitation abilityfor the objective of minimising the makespan. Marichelvam (2012) [7] developed an improved hybrid Cuckoo Search (IHCS) algorithmto solve PFFSP to minimise the makespan. The CS approach was improved by tuning the parameter values as the efficiency of the algorithm may decrease when the parameters of the Cuckoo Search are constant. Both the HCS and IHCS algorithms were implemented on some benchmark problems in the literature and the results showed both algorithms were better than other algorithms used in corresponding benchmark problems.

Job scheduling problem is combinatorial optimization problem which involves assignment of tasks to the machines in such a way that all task should get resource at time and complete their execution without affecting other tasks.Babukartik and Dhavachelvan (2012) [8] developed a hybrid algorithm consisting of ACO and CS algorithm for scheduling jobs. CS algorithm was used to improve the local search of the solution space efficiently using Lévy flights. Using a scheduler, the schedule was run and found that the makespan was lesser when using hybrid algorithm than that in ACO alone. An improved cuckoo search (ICS) algorithm was proposed by Selvi (2015) [9] combining two nature inspired metaheuristic algorithms, PSO and CS, for solving job scheduling problem. The advantage of creating a velocity value by PSO algorithm was used for generating new solutions which increases the efficiency of the improved approach in job scheduling and also in achieving the minimum makespan time. Computational results demonstrate that the ICS approach is a better method compared with the other simplex evolutionary algorithm. A hybrid algorithm combining the merits of genetic and cuckoo search algorithm was developed by Satyendra et al. (2015) [10] for a job scheduling problem for minimising makespan. The disadvantage of GA being trapped in local optima was overcome by using CS algorithm along with GA. Result analysis showed that performance of proposed algorithm was better than genetic algorithm.

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Hybrid flow shop (HFS) scheduling is an extension of the flow shop is in which the jobs will have the flexibility to get processed by any of the identical machines at any of the flow stages and is an important genre of NP-hard problems and requires extensive computational resources to perform scheduling.Marichelvam et al. (2014) [11] developedan Improved Cuckoo Search (ICS) metaheuristic algorithmfor the HFS scheduling problems.NEH heuristic was used prior to the CS algorithm to obtain optimal or near optimal solutions rapidly. The algorithm was tested with a real world furniture manufacturing company consisting of 20 jobs and 5 stage with parallel identical machines and compared with that of other metaheuristic algorithms.The computational results and comparison studies showed that ICS algorithm outperforms GA, ACO, PSO, and NEH heuristic in terms of average makespan obtained and also in terms of the best makespan obtained. An improved hybrid ACO Cuckoo algorithm was proposed by Ravianandan and Omkumar (2015) [12] to solve HFS problems. The proposed algorithm was being applied on a metal spinning industry consisting of 10 jobs and 5 stages. In this improved hybrid algorithm, the authors have shown that the CS algorithm was effectively useful for the assignment of jobs to machines and the ACO algorithm was helpful in providing the best job sequence corresponding to minimum makespan function.

In order to solve an assembly scheduling problem, Cuckoo Search (CS) algorithm was employed by Komaki et al. (2015) [13]. The paper deals with a Hybrid Flow Shop (HFS) scheduling problem followed by assembly stage was addressed. In the considered problem, there was S stage in HFS with identical parallel machines and machines never breakdown. At the assembly stage also there were identical parallel machines which assemble the parts based on the assembly structure of the products. The author used CS algorithm inorder to find the sequence of products and their components so that the makespan is minimized. The experimental results showed that in all cases SA was dominated by CS in terms of both quality and computational time.

Job shop scheduling problem is recognised as one of the most challenging NP-hard problems. Hanoun et al. (2012) [14] presented a Pareto Archived Multiobjective Cuckoo Search (PAMOCS) approach for solving the multiobjective job shop scheduling problem. The multi objective involves scheduling jobs on a single machine for minimising the total material waste as well as the total tardiness time. The solution accuracy of PAMOCS was compared with the closeness of the obtained solutions to the true Pareto front generated by the complete enumeration method. The computational results showed that the above approach was very effective and promising technique to solve job shop scheduling problems.

Scheduling of jobs is asignificant decision making task in aflexible manufacturing system (FMS) because it plays a vital role in deciding efficiency and capability of thesystem. ACS-based approach was developed by Burnwal and Deb (2013) [15] for scheduling optimization of a flexible manufacturing system. A Combined Objective Function (COF) was considered which involves minimization of total penalty cost due to delay (due date reduction) and maximizing the machine utilization time (idle time reduction). The proposed CS-based approach was demonstrated with slight modification in its Lévy flight inorder to accommodate the discrete nature of the solution on a standard FMS scheduling problem containing 43 jobs and 16 machines. The result comparison with GA and PSO showed that CS-based scheme provide better results obtained from the aforementioned methods.

IV. CONCLUSIONS AND FUTURE DIRECTIONS

From the above discussions it can be concluded that scheduling problems plays a vital role in industrial processes for improving the exploitation of resources, and therefore it is crucial to develop effective scheduling technologies. Cuckoo Search algorithm approach in scheduling optimization have proved to be very promising as it outperforms other evolutionary metaheuristic algorithms like GA, PSO, ACO

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etc. CS have the capability to do both local search and global search which makes them more efficient in exploring the search space on the global scale. Though there are some variant approaches of CS applied in scheduling problems, yet its potential has so far remainedunexplored in solving the scheduling optimization, which is an important and widely researched engineering optimization problem.

So far it has been identified that most of the scheduling problems used CS algorithm solely for the purpose of minimising the makespan criterion. Minimisation of total flow time, mean flow time, total tardiness and number of tardy jobs are some of the other important objective functions of scheduling problems which can be explored in the near future. Scheduling problems with setup and transportation times, machine breakdown, pre-emptions etc. may be another possible scope. All these challenging concerns may encourage more research in the near future.

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